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ARTIFICIAL INTELLIGENCE FOR ROBOTICS 2

Warehouse Robotics – Conveyor Belt Dynamics

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1 Introduction

This report focuses on the modelling of a warehouse logistics scenario using the Planning Domain Definition Language and PDDL+.

The core of this project is the orchestration of mobile manipulator robots operating in close coordination with a network of autonomous conveyor belts to transport packages from initial storage locations to designated delivery stations. In these environments, robots are responsible for manipulation tasks such as picking, loading, unloading, and delivering packages, while conveyor belts provide autonomous transportation between locations.

The objective is to analyze how different planning paradigms can represent environments in which both discrete robotic actions and autonomous processes coexist. In particular, the work investigates how conveyor belts can be approximated through discrete transitions in classical PDDL and later represented more realistically through continuous processes and events in PDDL+.

2 Domain Characteristics & Modelling Guidelines

The warehouse environment considered in this assignment is composed of:

- specific **interaction points** such as storage areas, load, unload, delivery zones and intermediate locations;
- generic **packages**;
- **mobile manipulators**. These, depending on the scenario, can be one or more than one, and can either navigate through the environment or operate within assigned manipulation zones to interact with packages and conveyor segments.
- **conveyor belts**, connecting different parts of the map, transporting packages across predefined paths

A key modelling guideline throughout the project is the explicit distinction between robot-controlled actions and autonomous conveyor dynamics. In the classical PDDL formulation, conveyor movement cannot be represented as a truly autonomous continuous phenomenon; therefore, it is approximated using discrete transitions through dedicated planning actions (conveyor advancement from one segment to another).

In the PDDL+ formulation, conveyor belts are modelled as autonomous systems through continuous processes.

3 Q1 – Basic PDDL Model

In the first part of the assignment, the warehouse scenario is modelled using classical PDDL. Since standard PDDL does not provide native support for continuous autonomous dynamics, the behavior of conveyor belts is approximated through a sequence of discrete transitions. The main objective of this model is therefore to represent the interaction between robots and conveyors while maintaining a fully discrete planning framework.

In this formulation, a clear distinction is maintained between:

- robot-controlled actions, such as navigation, picking, loading, retrieving, and delivery;
- conveyor dynamics, approximated through explicit planning actions that simulate the advancement of a package along belt segments.

The domain has been designed to support both situations in which conveyor usage is simply advantageous and scenarios in which conveyors become strictly necessary to reach otherwise inaccessible areas of the environment.

3.1 Domain Modelling

The domain is defined through three main object types:

- robot: mobile manipulators operating in the warehouse;
- package: items that must be transported;
- location: storage cells, intermediate nodes, conveyor segments, and delivery stations.

Predicates

The predicate set is organized to separately model navigation, manipulation, and conveyor interactions.

Navigation predicates

- **connected(?from ?to)**, represents connections between locations for robot movement;
- **robot-at(?r ?l)**, specifies the current position of a robot.

Package Manipulation predicates

- **package-at(?p ?l)**, indicates the current position of a package;
- **holding(?r ?p)**, states that a robot is carrying a package;
- **handempty(?r)**, ensures that a robot can manipulate only one package at a time.

Conveyor-related predicates

- **belt-connected(?from ?to)**, defines adjacency relations between conveyor belt segments;
- **on-belt(?p)**, indicates that a package is currently being transported by a conveyor;
- **belt-free(?l)**, represents whether a conveyor segment is available for a package.

Interaction zones

- **load-zone(?l)**, locations where robots can place packages onto conveyors;
- **unload-zone(?l)**, locations where packages can be retrieved from conveyors;
- **delivery-zone(?l)**, final destinations for package delivery.

3.2 Action Modelling

The domain includes six actions that model both robot operations and conveyor advancement.

The **move** action allows a robot to travel between adjacent connected locations. Each movement increases the global execution cost.

For the package manipulation:

- **pick**: allows a robot to collect a package from an unloading zone;
- **place**: deposits a carried package inside a delivery zone.

The robot with the **load-conveyor** action can place a package onto a conveyor entry segment and with **retrieve** action removes a package from the conveyor at an unloading point. These are the actions that define the interaction boundary between robots and autonomous transportation infrastructure.

The action **conveyor-step** is the core mechanism used to approximate conveyor dynamics in classical PDDL. Instead of modelling continuous movement, the package progresses discretely from one belt segment to the next through planner-generated actions.

3.3 Optional Conveyor Environment

In the first problem configuration, conveyor usage is optional. The warehouse topology still allows the robot to manually transport packages to their destinations using only navigation and manipulation actions. However, conveyor belts provide a potentially cheaper and more efficient transportation alternative.

As shown in Figure 1, the environment contains:

- one mobile robot (r1);
- two packages (pck1, pck2);
- two conveyor systems:
 1. a top conveyor connecting I2 → I3 → I4;
 2. a bottom conveyor connecting I9 → I10 → I11 → I12.

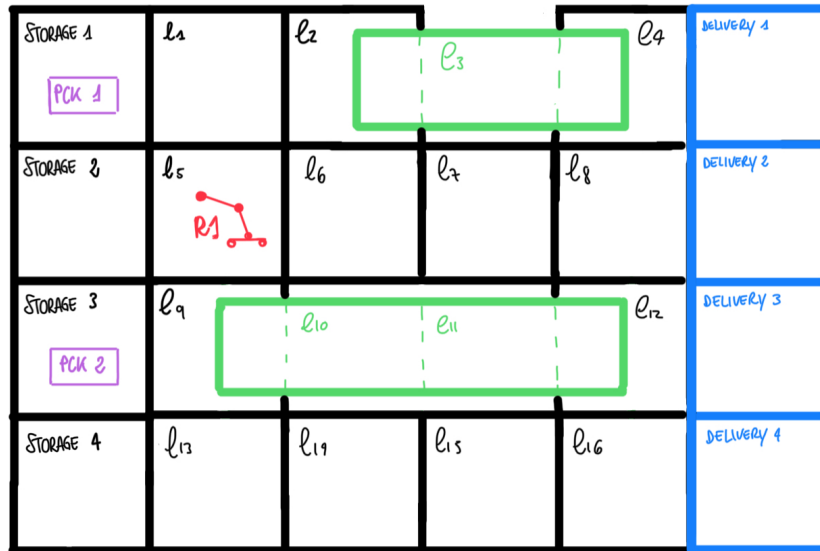


Figure 1: Environment with optional use of Conveyor Belts

The robot starts in the central area of the warehouse, while packages are initially positioned in storage1 and storage3. The goal requires delivering: pck1 to delivery1 and pck2 to delivery4.

Since standard navigational paths still exist between all relevant areas, the planner may choose whether using the conveyor system is beneficial with respect to the total-cost metric.

The resulting model demonstrates how classical PDDL can encode hybrid transportation strategies combining direct robot transportation and conveyor-assisted movement.

Results Analysis – Optional Conveyor Environment

In this scenario, the warehouse topology still allows the robot to reach all relevant storage and delivery locations through standard navigation. Therefore, the conveyor belts are not strictly required to solve the task, but they may provide a cheaper alternative. Two different plans were obtained for this scenario.

The first solution has:

```

| move r1 l5 l9
| move r1 l9 storage3
| pick pck2 r1 storage3
| move r1 storage3 l9
| move r1 l9 l5
| move r1 l5 l6
| move r1 l6 l7
| move r1 l7 l8
| move r1 l8 delivery2
| move r1 delivery2 delivery3
| move r1 delivery3 delivery4
| place pck2 r1 delivery4
| move r1 delivery4 delivery3
| move r1 delivery3 delivery2
| move r1 delivery2 l8
| move r1 l8 l7
| move r1 l7 l6
| move r1 l6 l5
| move r1 l5 l1
| move r1 l1 storage1
| pick pck1 r1 storage1
| move r1 storage1 l1
| move r1 l1 l5
| move r1 l5 l6
| move r1 l6 l7
| move r1 l7 l8
| move r1 l8 l4
| move r1 l4 delivery1
| place pck1 r1 delivery1

```

```

Plan-Length:29
Metric (Search):54.0
Planning Time (msec): 40
Heuristic Time (msec): 18
Search Time (msec): 37
Expanded Nodes:41
States Evaluated:68
Number of Dead-Ends detected:3
Number of Duplicates detected:52

```

Figure 3: Costs Optional-NonOptimal Environment

Figure 2: Plan Optional-NonOptimal Environment

This solution is very fast to compute because the planner follows a direct and simple strategy. The search space is small, as shown by the low number of expanded nodes and evaluated states. However, the resulting cost is not the best one, because the robot performs many navigation actions while physically carrying the packages.

The second solution for the optional environment has:

```

| move r1 l5 l9
| move r1 l9 l5
| move r1 l5 storage2
| move r1 storage2 storage1
| pick pck1 r1 storage1
| move r1 storage1 storage2
| move r1 storage2 l5
| move r1 l5 l9
| load-conveyor r1 pck1 l9
| conveyor-step pck1 l9 l10
| conveyor-step pck1 l10 l11
| move r1 l9 storage3
| pick pck2 r1 storage3
| move r1 storage3 l9
| move r1 l9 l13
| move r1 l13 l14
| move r1 l14 l15
| move r1 l15 l16
| move r1 l16 delivery4
| place pck2 r1 delivery4
| move r1 delivery4 delivery3
| move r1 delivery3 l12
| conveyor-step pck1 l11 l12
| retrieve r1 pck1 l12
| move r1 l12 l8
| move r1 l8 l4
| move r1 l4 delivery1
| place pck1 r1 delivery1

```

Figure 4: Plan Optional-Optimal Environment

This plan is slightly shorter and has a significantly lower metric cost. Unlike the previous one, the planner decides to exploit the conveyor system for part of the package transportation.

This result shows that the conveyor can be useful even in an environment where it is not strictly required for reachability. By using the conveyor, the robot can reduce part of its manual navigation and therefore obtain a lower total execution cost.

However, this optimal solution is not immediately found by ENHSP in its standard configuration. The reason is that ENHSP is mainly a heuristic planner: it is designed to find a feasible plan efficiently, rather than exhaustively proving that the first plan found is cost-optimal. As a consequence, the planner may stop as soon as it finds a valid solution, even if another solution with a lower metric cost exists.

3.4 Necessary Conveyor Environment

The second problem configuration introduces structural constraints that make conveyor usage mandatory.

Compared to the previous scenario, as the Figure 6 show, several direct navigational connections are intentionally removed from the warehouse graph. As a consequence, some areas become unreachable through robot navigation alone. Conveyor belts therefore act as mandatory transportation bridges between disconnected regions.

With this scenario there are two robots, Robot r1 operates primarily in the left side of the warehouse, while Robot r2 operates closer to the delivery area.

The planner must therefore coordinate robot actions with conveyor usage:

1. packages are collected from storage areas;
2. loaded onto conveyor entry points;

```

Plan-Length:28
Metric (Search):44.0
Planning Time (msec): 209
Heuristic Time (msec): 156
Search Time (msec): 208
Expanded Nodes:3934
States Evaluated:4468
Number of Dead-Ends detected:1301
Number of Duplicates detected:7219

```

Figure 5: Costs Optional-Optimal Environment

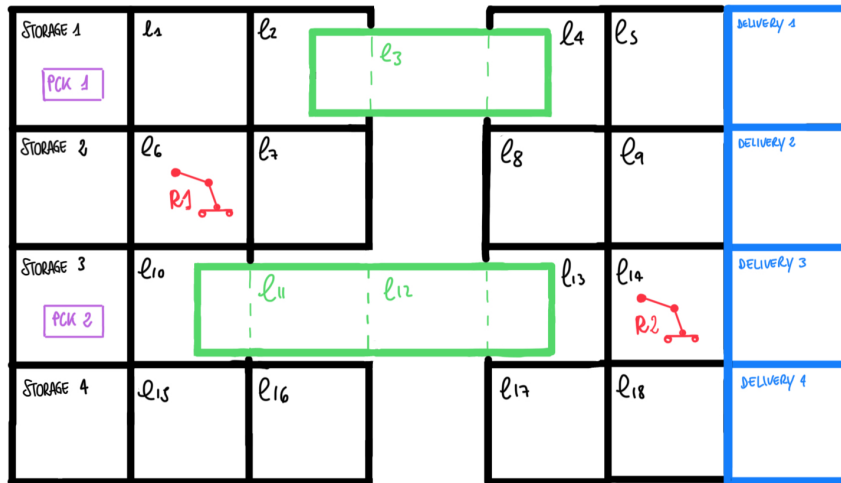


Figure 6: Environment with necessary use of Conveyor Belts

3. advanced along conveyor segments through conveyor-step;
4. retrieved at unloading zones;
5. finally delivered to their destinations.

Unlike the optional environment, here the conveyor system is not simply an optimization mechanism but an essential component for achieving reachability and satisfying the planning goals.

Results Analysis – Necessary Conveyor Environment

The first solution obtained for this scenario has:

```

| move r1 l6 l10
| move r2 l14 l9
| move r2 l9 l5
| move r1 l10 storage3
| move r2 l5 l4
| pick pck2 r1 storage3
| move r1 storage3 l10
| load-conveyor r1 pck2 l10
| move r1 l10 l6
| move r1 l6 l1
| move r1 l1 storage1
| pick pck1 r1 storage1
| move r1 storage1 l1
| move r1 l1 l2
| load-conveyor r1 pck1 l2
| move r1 l2 l1
| conveyor-step pck2 l10 l11
| conveyor-step pck2 l11 l12
| conveyor-step pck2 l12 l13
| move r2 l4 l8
| move r1 l1 storage1
| move r2 l8 l13
| retrieve r2 pck2 l13
| move r2 l13 l14
| move r2 l14 delivery3
| move r1 storage1 l1
| move r1 l1 l2
| conveyor-step pck1 l2 l3
| conveyor-step pck1 l3 l4
| move r2 delivery3 delivery4
| place pck2 r2 delivery4
| move r2 delivery4 delivery3
| move r2 delivery3 delivery2
| move r2 delivery2 delivery1
| move r2 delivery1 l5
| move r2 l5 l4
| retrieve r2 pck1 l4
| move r2 l4 l5
| move r2 l5 delivery1
| place pck1 r2 delivery1

```

```

Plan-Length:38
Metric (Search):58.0
Planning Time (msec): 193
Heuristic Time (msec): 134
Search Time (msec): 192
Expanded Nodes:2100
States Evaluated:3644
Number of Dead-Ends detected:95
Number of Duplicates detected:10595

```

Figure 8: Costs Necessary-NonOptimal Environment

Figure 7: Plan Necessary-NonOptimal Environment

Robot r1 picks packages from the storage areas and loads them onto the conveyor. Then, the packages are advanced along the belt through conveyor-step actions. Robot r2, operating on the right side of the environment, retrieves the packages from the unloading area and places them in the correct delivery zones. Without the conveyor, the packages could not be transferred between these two regions.

A second, more optimized solution was also obtained:

```

| move r1 l6 l10
| move r2 l14 l13
| move r1 l10 storage3
| move r2 l13 l8
| move r2 l8 l4
| pick pck2 r1 storage3
| move r1 storage3 l10
| load-conveyor r1 pck2 l10
| move r1 l10 l6
| move r1 l6 l1
| move r1 l1 storage1
| pick pck1 r1 storage1
| move r1 storage1 l1
| move r1 l1 l2
| load-conveyor r1 pck1 l2
| conveyor-step pck1 l2 l3
| conveyor-step pck1 l3 l4
| conveyor-step pck2 l10 l11
| retrieve r2 pck1 l4
| move r2 l4 l5
| move r2 l5 delivery1
| place pck1 r2 delivery1
| move r2 delivery1 delivery2
| move r2 delivery2 delivery3
| move r2 delivery3 l14
| move r2 l14 l13
| conveyor-step pck2 l11 l12
| conveyor-step pck2 l12 l13
| retrieve r2 pck2 l13
| move r2 l13 l17
| move r2 l17 l18
| move r2 l18 delivery4
| place pck2 r2 delivery4

```

Figure 9: Plan Necessary-Optimal Environment

This plan improves both the plan length and the metric cost. The cost decreases from 58.0 to 48.0, and the plan length decreases from 38 to 33 actions. The planner achieves this by choosing a better ordering of robot movements, conveyor loading, conveyor advancement, and retrieval actions.

However, this improvement requires a much larger computational effort. The number of expanded nodes increases from 2100 to 30087, and the number of dead-ends increases dramatically.

```

Plan-Length:33
Metric (Search):48.0
Planning Time (msec): 1045
Heuristic Time (msec): 852
Search Time (msec): 1041
Expanded Nodes:30887
States Evaluated:32867
Number of Dead-Ends detected:25827
Number of Duplicates detected:142569

```

Figure 10: Costs Necessary-Optimal Environment

4 Q2 – Advanced PDDL+ Model

In the second part of the assignment, the warehouse system is extended using PDDL+. Instead of generating conveyor movements as discrete actions, the PDDL+ model introduces continuous processes and instantaneous events. This allows the packages to move automatically along the conveyor belts once they are placed on them, while the robots are only responsible for manipulation, interception, and delivery operations.

4.1 Domain Modelling

The PDDL+ domain is based on four main object types:

- **robot**, representing the fixed manipulators operating at specific interaction zones;
- **package**, representing the items transported;
- **location**, representing conveyor segments, loading zones, unloading zones, storage points, and delivery stations;
- **conveyor**, representing the autonomous conveyor belts.

The environment is composed of five conveyor systems: cMain, cRight, cLeft, cBackR, and cBackL. The main conveyor connects the central storage area, while the right and left conveyors transport packages toward different delivery-side robots. The back conveyors close the loop by bringing packages back toward the main storage area when needed.

The most important predicates used in the domain are:

- **belt-segment(?c ?from ?to)**, maps the layout of a specific conveyor belt;
- **conveyor-entry(?c ?l)** / **conveyor-exit(?c ?l)**, which identify the entry and exit points of each belt;
- **conveyor-connected(?c1 ?c2 ?l1 ?l2)**, which allows a package to automatically pass from one conveyor to another;
- **robot-act-on(?r ?l)**, which restricts each robot to the locations it can physically reach, they are not mobile agents;
- **belt-free(?l)**, which prevents two packages from occupying the same conveyor segment;
- **busy(?r)**, which models whether a robot is currently performing an operation.

4.2 Numeric Fluents and Continuous Dynamics

The PDDL+ model introduces numeric fluents to represent time-dependent quantities:

- **(belt-progress ?p)**, which represents the progress of a package along its current conveyor segment;
- **(conveyor-speed ?c)**, which defines the speed of each conveyor;
- **(busy-timer ?r)**, which represents the remaining time before a robot becomes available again;
- **(total-cost)**, which is minimized by the planner.

4.3 Actions, Processes and Events

The domain separates three different types of dynamics. The **Discrete Actions** represent intentional robot operations:

- **load-conveyor**, used by robot1 to place a package onto a conveyor entry point;
- **intercept**, used by intermediate robots (robot 2 to 7) to catch a package while it is moving on a conveyor segment;
- **grasp**, used to retrieve a package from the final segment of a conveyor (used by robot1 at storage2);
- **deliver**, used to place a held package into a delivery zone;

- **manipulate-with-package / manipulate-no-package**, used when a robot changes its working configuration while holding a package or without carrying a package. These actions carry a heavy execution cost (15) and a long duration (2.0 seconds) to avoid useless motion of the robots;

The **Continuous Processes** represent autonomous background dynamics:

- **package-movement**, which moves packages along conveyor belts. This process continuously increases the progress of a package while it is on a running conveyor:

$$\Delta \text{belt-progress}(p) = \#t \times \text{conveyor-speed}(c)$$

This process is active as long as the package is on a running conveyor and its progress is less than 1.0. Therefore, once a package is on a belt, it moves automatically over time.

- **robot-timers**, decreases the busy timer of each robot. This models the fact that manipulation operations are not instantaneous: after grasping, delivering, or changing configuration, a robot remains occupied for a certain amount of time before it can execute another action;

The **Instantaneous Events** represent instantaneous state transitions triggered by numeric conditions:

- **advance-segment**, activated when a package reaches a progress ≥ 1.0 on an internal segment. It clears the previous segment, occupies the next one, and resets the progress fluent to 0.0.
- **reach-exit**, activated when a package reaches the end of a conveyor belt handling the automatic physical transition of a package between two connected conveyor belts (e.g., from `cRight` to `cBackR`)
- **free-robot**, automatically resets a robot's `busy` state back to false the moment its `busy-timer` reaches 0.0, freeing its resources for the next action.

4.4 Problem 1: Two-Package Scenario

The first PDDL+ problem considers a relatively simple configuration with two packages. The environment includes seven robots and five conveyor belts operating at a constant speed of 0.8 units/sec.

The initial state defines a sparse distribution:

- **p1** starts on `cRight` at location 14;
- **p2** starts on `cMain` at location 12.

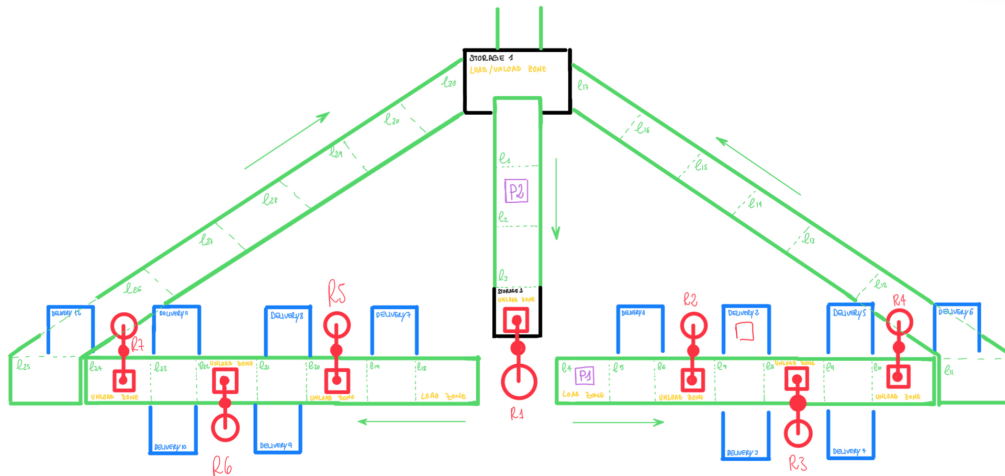


Figure 11: Two-package Scenario

The goal requires delivering `p1` to `delivery2` and `p2` to `delivery12`. This scenario tests whether the planner can coordinate the continuous movement of the conveyors with the manipulation actions of the robots.

The resulting plan has:

```

Found Plan:
0: ----waiting---- [4.0]
4.0: (intercept r2 p1 l6 cRight)
4.0: ----waiting---- [5.0]
5.0: (manipulate-with-package r2 p1 l6 delivery2)
5.0: ----waiting---- [6.0]
6.0: (grasp r1 p2 storage2 cMain)
6.0: ----waiting---- [7.0]
7.0: (manipulate-with-package r1 p2 storage2 l18)
7.0: (deliver r2 p1 delivery2)
7.0: ----waiting---- [9.0]
9.0: (load-conveyor r1 p2 l18 cLeft)
9.0: ----waiting---- [21.0]
21.0: (grasp r7 p2 l24 cLeft)
21.0: ----waiting---- [22.0]
22.0: (manipulate-with-package r7 p2 l24 delivery12)
22.0: ----waiting---- [24.0]
24.0: (deliver r7 p2 delivery12)

```

Figure 12: Plan two-package Scenario

```

Plan-Length:43
Elapsed Time: 24.0
Metric (Search):75.0
Planning Time (msec): 52
Heuristic Time (msec): 34
Search Time (msec): 46
Expanded Nodes:34
States Evaluated:52
Number of Dead-Ends detected:4
Number of Duplicates detected:1

```

Figure 13: Costs two-package Scenario

The plan shows that the planner is able to solve the task efficiently. Package p1 is intercepted from the right conveyor by robot r2, manipulated to the correct delivery configuration, and delivered to delivery2. Package p2 instead follows a longer path: it reaches storage2, is grasped by r1, transferred to the left conveyor, and later intercepted by r7, which delivers it to delivery12.

The low number of expanded nodes and evaluated states indicates that the search space is still limited in this configuration. Since only two packages are active, the planner has fewer possible conflicts to resolve. The conveyor dynamics still introduce temporal constraints, but the number of interactions between packages is small.

4.5 Problem 2: Four-Package Scenario

The second PDDL+ problem increases the complexity of the task by introducing four packages at the same time:

- p1 and p2 are closely queued on cRight (l4 and l5);
- p3 and p4 are placed sequentially on cMain (l11 and l2).

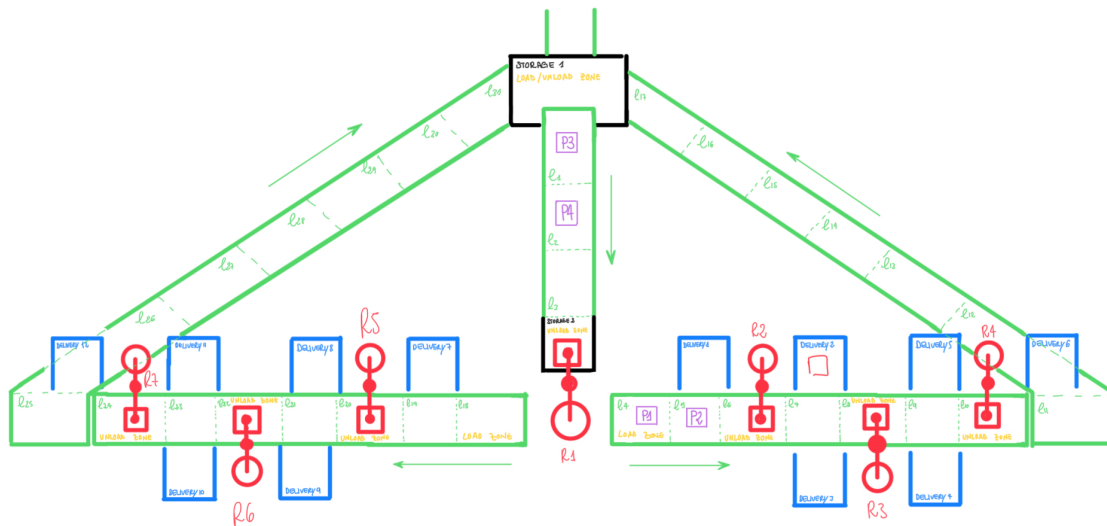


Figure 14: Four-package Scenario

The objective is to route and deliver p1 and p2 at delivery2, p3 at delivery10, and p4 at delivery8. The resulting plan has:

```

0: ----waiting---- [5.0]
5.0: (intercept r2 p1 l6 cRight)
5.0: ----waiting---- [7.0]
7.0: (grasp r1 p4 storage2 cMain)
7.0: ----waiting---- [8.0]
8.0: (manipulate-with-package r1 p4 storage2 l18)
8.0: ----waiting---- [10.0]
10.0: (load-conveyor r1 p4 l18 cLeft)
10.0: (manipulate-no-package r1 l18 storage2)
10.0: ----waiting---- [13.0]
13.0: (grasp r1 p3 storage2 cMain)
13.0: ----waiting---- [14.0]
14.0: (manipulate-with-package r1 p3 storage2 l18)
14.0: (intercept r5 p4 l20 cLeft)
14.0: ----waiting---- [15.0]
15.0: (manipulate-with-package r5 p4 l20 delivery8)
15.0: ----waiting---- [17.0]
17.0: (deliver r5 p4 delivery8)
17.0: ----waiting---- [34.0]
34.0: (load-conveyor r1 p3 l18 cLeft)
34.0: (manipulate-no-package r1 l18 storage2)
34.0: ----waiting---- [36.0]
36.0: (grasp r1 p2 storage2 cMain)
36.0: ----waiting---- [37.0]
37.0: (manipulate-with-package r1 p2 storage2 l4)
37.0: (manipulate-with-package r2 p1 l6 delivery2)
37.0: ----waiting---- [39.0]
39.0: (load-conveyor r1 p2 l4 cRight)
39.0: (manipulate-no-package r5 delivery8 l20)
39.0: ----waiting---- [43.0]
43.0: (intercept r6 p3 l22 cLeft)
43.0: ----waiting---- [44.0]
44.0: (manipulate-with-package r6 p3 l22 delivery10)
44.0: (deliver r2 p1 delivery2)
44.0: (manipulate-no-package r2 delivery2 l6)
44.0: ----waiting---- [46.0]
46.0: (deliver r6 p3 delivery10)
46.0: ----waiting---- [75.0]
75.0: (manipulate-no-package r1 l4 storage2)
75.0: ----waiting---- [77.0]
77.0: (grasp r1 p2 storage2 cMain)
77.0: ----waiting---- [78.0]
78.0: (manipulate-with-package r1 p2 storage2 l4)
78.0: ----waiting---- [80.0]
80.0: (load-conveyor r1 p2 l4 cRight)
80.0: ----waiting---- [84.0]
84.0: (intercept r2 p2 l6 cRight)
84.0: ----waiting---- [85.0]
85.0: (manipulate-with-package r2 p2 l6 delivery2)
85.0: ----waiting---- [87.0]
87.0: (deliver r2 p2 delivery2)

```

Figure 15: Plan four-package scenario

```

Plan-Length:171
Elapsed Time: 87.0
Metric (Search):298.0
Planning Time (msec): 177759
Heuristic Time (msec): 154477
Search Time (msec): 177754
Expanded Nodes:1315680
States Evaluated:1724374
Number of Dead-Ends detected:561267
Number of Duplicates detected:1891206

```

Figure 16: Costs four-package scenario

Compared with the two-package scenario, the plan is much longer and the computational effort increases dramatically. The plan length grows from 43 to 171 actions, while the number of expanded nodes increases from only 34 to more than 1.3 million.

The main reason for this increase is the interaction between continuous conveyor movement and discrete robot availability. If a robot is busy or in the wrong configuration or takes too long to intercept or grasp an item, the package may continue moving.

The comparison between the two scenarios highlights the effect of increasing the number of packages in a PDDL+ model.

In the two-package case, the planner finds a solution quickly. The system has enough free space on the conveyors, and each package can be handled independently by the assigned robots.

In the four-package case, the same domain becomes much more computationally demanding. The reason is not only the larger number of packages, but also the stronger coupling between them. Packages are not independent anymore: the motion of one package can block or delay another package.

5 Discussion

The two modelling approaches developed in this assignment highlight the difference between agent-controlled dynamics and environment-driven dynamics. In the basic PDDL model, every relevant change in the world must be explicitly represented as a discrete action selected by the planner.

This is the main approximation introduced in Q1. Each time a package has to move from one conveyor segment to the next one, the planner must explicitly insert a specific action in the plan. Therefore, the conveyor is not truly autonomous: it only moves when the planner decides to apply an action to a package.

This creates an important limitation. In a real warehouse, if a package is placed on a moving conveyor belt, it starts moving automatically. If another package is placed on the same conveyor after it, both packages

should move according to the belt motion, while respecting spacing and occupancy constraints. In the classical PDDL model, instead, each package must be advanced separately. For example, if one package is moved forward with conveyor-step, the other package does not move automatically. The planner has to generate another conveyor-step action for the second package. This makes the model less realistic.

Another limitation of the discrete approximation is the loss of temporal information. The PDDL model does not represent the real speed of the conveyor, the time needed for a package to reach a transfer point, or the exact moment at which a robot should be ready to retrieve or intercept it. As a consequence, the model can only describe the order of actions, but not the continuous evolution of the system over time. This can hide important synchronization problems that would appear in a real automated warehouse.

The discretization of continuous transport can also lead to unrealistic behaviours. Since package movement is represented as a sequence of instantaneous transitions, the model does not capture intermediate positions along the conveyor. A package is either on one segment or on the next one, with no representation of the continuous motion between them. This may cause a loss of physical realism, especially when robots are supposed to intercept moving packages. In reality, a robot must anticipate the position of the package and act at the correct time. In the basic PDDL model, this anticipation is simplified into reaching the correct discrete location before applying the retrieval action.

The results obtained with ENHSP also show an important aspect of heuristic planning. In some cases, ENHSP does not immediately return the optimal plan, but rather the first feasible plan found by the search. This is especially visible in the optional and necessary conveyor environment with PDDL. The planner can solve the problem by using only robot navigation, because the conveyor is not required for reachability. This direct solution is easier to find because it follows a simple strategy: the robot picks a package, moves through connected locations, and delivers it. Nevertheless, it is harder for the planner to discover the optimal plan, because it requires additional actions and a less direct sequence: the robot must load the package onto the conveyor, apply conveyor-step actions, retrieve the package at the unloading point, and then complete the delivery. For this reason, the optimal conveyor-assisted plan requires a larger search effort, with more expanded nodes, more evaluated states, and more dead-ends. A heuristic planner may prefer a direct and easily reachable solution, even if another plan with a lower cost exists. To obtain the optimal plan, the planner must explore a larger part of the search space and compare different strategies.

The PDDL+ model addresses many of these limitations. In Q2, conveyor motion is represented as an autonomous continuous process. Once a package is placed on a conveyor belt, its progress increases over time according to the conveyor speed. The planner no longer decides every single movement of the package. Instead, it must plan robot actions while anticipating the autonomous evolution of the environment.

This makes the model more realistic. Conveyor belts become environment-driven systems, while robots remain agent-controlled entities. The planner must reason about where a package will be in the future, when it will reach a transfer point, and which robot must be ready to intercept or retrieve it. Therefore, PDDL+ better captures the synchronization problem between robots and autonomous infrastructure.

However, this increased realism also makes the planning problem more complex. The planner must handle both discrete actions and continuous numeric change. In dense scenarios with multiple packages, it must also avoid conflicts between packages sharing the same conveyor segments. This explains why the four-package PDDL+ problem produces a much larger search space than the simpler two-package scenario.

Overall, the comparison between PDDL and PDDL+ shows that classical PDDL is useful for validating the logical structure of the task, such as loading, unloading, and delivery sequences. However, it remains limited when modelling autonomous continuous transport. PDDL+ is more suitable for representing real conveyor belt dynamics, because it separates robot-controlled actions from environment-driven processes and forces the planner to anticipate the autonomous motion of packages.